2023

PHYSICS — HONOURS — PRACTICAL

Paper : SEC-1P

Full Marks : 100

The figures in the margin indicate full marks.

SET - 1

[LNB - 20, VIVA - 20, PROGRAMMING - 60]

Answer ***any one*** question from ***each group.***

Group - A

1. Three functions are defined in the given range as the following :

$$f(x) = 1.45x^3 - 6\sqrt{x}, \quad x \geq 2$$
$$g(x) = \sin^2(x), -1 \leq x < 2$$
$$h(x) = |x|e^x, \quad x < -1$$

Set the range of x values from –2 to +3 and the range of y values from –5 to +5. Set ticks of the x-axis at an interval of 1.0 and ticks of y-axis at an interval 2.0. Set title 'Piece-wise continuous functions'. Plot the three functions using GNUPLOT and store the plot in suitable format. 20

2. Plot the following functions for $0 \leq x \leq \pi$ on the same graph using GNUPLOT. Set 1000 samples. Mark each curve with different line style and line widths. Label x-axis and y-axis as 'x-values' and 'y-values' respectively. Set the title of the plot 'Function plots'. Save the plot in a suitable format.

$$f_1(x) = \sin^2\left(\sqrt{x}\right), \quad f_2(x) = e^{-x/2}\cos(2\pi x), \quad f_3(x) = 12.34x^{5/3}$$

20

3. Plot the following functions for $-10 \leq x \leq 10$ on the same graph using GNUPLOT. Set 1000 samples. Take the range of y-values from -4 to $+4$. Mark each curve with different line style and line widths. Label x-axis and y-axis as 'x-values' and 'y-values' respectively. Set the title of the plot 'Function plots'. Save the plot in a suitable format.

$$f_1(x) = \coth(x) - \frac{1}{x}, \quad f_2(x) = e^{-\sin(x)}, \quad f_3(x) = x|x|$$

20

4. Plot the following parametric functions using GNUPLOT :

$$x(r, t) = r(t)\cos(t), \quad y(r, t) = r(t)\sin(t),$$

where $r(t) = 3(1 + \cos(t))$. Set the range of t values from $-\pi$ to π. Set the title of the plot as 'Cardioid function'. Set samples 1000. Save the plot in a suitable format. 20

5. Plot the following polar equations in GNUPLOT :

$$r(\theta) = \theta^2 \sin(\theta/\pi), \quad r(\theta) = 3.75e^{-\frac{\theta}{2.39}}$$

Consider θ range to be $-\pi$ to π and the radial (r) range to be $-3 < r < 3$. Set different line types for the two functions. Put the title "Polar function plot". Save the plot in a suitable format. 20

6. Save the following data table in a file. Fit the data points of the second row against that of

x	−3.0	−2.0	−1.0	0.0	1.0	2.0	3.0	4.0
$f(x)$	3.422	4.057	4.808	5.7	6.756	8.008	9.492	11.251

the first row using GNUPLOT with the equation $f(x) = ae^{bx}$. Write the values of the fitted parameters a, b. Show the fitted curve along with the data points in the same plot. Set the title "Data fitting" and 'x' and '$f(x)$' for x-label and y-label respectively. Save the plot in a suitable format. 20

7. Create the following data table in a file :

x	0.0	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0
y	0.0	0.006	0.053	0.195	0.49	1.0	1.792	2.935	4.5	6.56	9.19

Fit the above data table with $f(x) = x^a$ with parameter a. Write the value of a. Plot the above data table along with $f(x)$.

Set the title 'Data fitting' and 'x' and '$f(x)$' for x-label and y-label respectively. Save the plot in a suitable format. 20

Group - B

8. (a) Write a python program to find the root of the transcendental equation correct up to three decimal places $\sin(x) = \ln(x)$ in between $[0.001, \pi]$ using Bisection method.

   (b) Write a python program to calculate the sum of first 20 natural numbers. Print the output. 30+10

9. (a) Write a python program to find the velocity of an object at $t = 10$s by solving the following differential equation using Euler method :

$$\frac{dv}{dt} = 3t^2 - \sqrt{t} - \cos(6t).$$

The initial velocity of the object is given by $v(t = 0) = 5$ m / s.

Create a list of t-values with 1000 elements between [0, 10]. Put the y-values in a list Y. Print the last element of Y.

(b) Consider the following matrix :

$$A = \begin{pmatrix} 1 & 4 & -2 \\ 2 & -1 & 3 \\ -4 & -3 & 1 \end{pmatrix}$$

Write a python program to find the transpose of matrix A. Print the new matrix. 30+10

10. (a) Using Newton-Raphson method with starting point $x_0 = 1$, determine the root of the following equation correct up to three decimal places :

$$x^3 - 2x^2 - 5x + 8 = 0.$$

Print the root.

(b) Write a python program to calculate the sum of the following G.P. series :

S = 2 + 6 + 18 + ... up to 5 terms.

Print the output. 30+10

11. (a) Using Newton-Raphson method with starting point $x_0 = 1$, determine the root of the following equation correct up to three decimal places :

$$x \cos x - x^2 = 0.$$

Print the root.

(b) Consider the following two matrices :

$$A = \begin{pmatrix} 3 & 4 \\ 5 & 6 \end{pmatrix}, \quad B = \begin{pmatrix} -1 & 0 \\ 1 & -2 \end{pmatrix}$$

Write a python program to find $A + B$. Print the resultant matrix. 30+10

12. (a) A first order differential equation is given as below :

$$\frac{dy}{dx} = 4x - 2 \quad \text{with } y(0) = 0.$$

Create a list (say X) with 100 x-values between [0, 1]. Solve this differential equation using Euler method for x-values in the list X. It is given that the exact solution is $y(x) = 2x^2 - 2$.

Calculate the y-value for the last element of the list X. Print this value and compare with the value from the exact solution.

(b) Create a list with following numbers :

27, 11, 5, 18.

Find the even and odd numbers and print accordingly. 30+10

13. (a) Write a python program to compute the exponential series up to three decimal places of accuracy :

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots$$

Take $x = 1$ as input. Print the value of $ln\,(e)$ using the computed value of e.

(b) Write a python program to calculate and print the trace of the following matrix :

$$\begin{pmatrix} -7 & 2 & 1 \\ -3 & 4 & 8 \\ 5 & -1 & 2 \end{pmatrix}$$

30+10

14. (a) Calculate the sum of the following convergent series for $n = 2$ correct up to 5 decimal places :

$$1 + \frac{1}{n} + \frac{1}{n^2} + \frac{1}{n^3} + \ldots$$

Verify that the sum is close to $\frac{n}{n-1}$.

(b) The sequence F_n of Fibonacci numbers is defined by the recurrence relation :

$F_n = F_{n-1} + F_{n-2}$ with seed values and $F_0 = 0$ and $F_1 = 1$. Calculate and print the 10th term of this series.

30+10

2023

PHYSICS — HONOURS — PRACTICAL

Paper : SEC-1P

Full Marks : 100

The figures in the margin indicate full marks.

SET - 2

[LNB - 20, VIVA - 20, PROGRAMMING - 60]

Answer ***any one*** question from ***each group.***

Group - A

1. Save the following data table in a file : Fit the data points of the second row against that of the

x	1.22	1.44	1.67	1.89	2.11	2.33	2.56	2.78	3.
$f(x)$	5.013	3.65	2.78	2.19	1.77	1.467	1.234	1.054	0.910

first row using GNUPLOT with the equation $f(x) = ax^b$. Write the values of the fitted parameters a, b. Show the fitted curve along with the data points in the same plot. Set the title "Data fitting" and 'x' and '$f(x)$' for x-axis and y-axis respectively. 20

2. Plot the following functions for $-1 \le x \le 3$ on the same graph using GNUPLOT. Set the range of y-values to be from –2 to 4. Set 1000 samples. Mark each curve with different line style and line widths. Label x-axis and y-axis as 'x-values' and 'y-values' respectively. Set the title of the plot 'Function plots'. Save the plot in a suitable format.

$$f_1(x) = \text{sech}\left(|x|\right),\ f_2(x) = |x|^3 \ln(x)/(4+\sqrt{x}),\ f_3(x) = 0.83x^3 - 4.2\,x^{1/4}$$

20

3. Show the polar plot of the function $-\frac{3}{2}\left(5\cos^2 t - 1\right)\sin t$, where t is the angle made with the reference directions (positive x-axis) using GNUPLOT. Set samples to 800. Set polar grid with appropriate angle between gridlines. Take $0 \le t \le 2\pi$ and $0 \le r \le 1$. Set title of the plot as "Third order associated Legendre polynomial". Save the plot in a suitable format. 20

4. Three functions are defined in the given range as the following :

$$f(x) = 0.71x^2 - 6.29\sqrt{x}, \quad x \geq 4$$
$$g(x) = \sin^2(x^3), \; 2 \leq x < 4$$
$$h(x) = e^{|x|}, \quad -2 \leq x < 2.$$

Set the range of x values from –2 to +5 and the range of y values from –2 to +10. Set ticks of the x-axis at an interval of 0.5 and ticks of y-axis at an interval 1.0. Set title "Piece-wise continuous functions". Plot the three functions using GNUPLOT and store the plot in a suitable format. 20

5. Plot the following polar functions in GNUPLOT :

$$r(\theta) = 2\cos(2\theta), \quad r(\theta) = \frac{\theta}{3}.$$

Take $0 \leq \theta \leq 2\pi$. Set legend to each function as "Rose petal" and "Spiral of Archimedes" respectively. Set title "Some polar functions". Use different line styles and line widths for the two curves. Save the plot in a suitable format. 20

6. Make the following data table in a file :

x	0.0	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0
y	0.0	0.366	0.424	0.425	0.402	0.368	0.33	0.292	0.255	0.222	0.191

Write a gnuplot script to fit the data with the curve $f(x) = \sqrt{x}e^{-ax}$ with parameter a. Write the value of a. Plot the above data table along with $f(x)$. Set the title "Data fitting" and 'x' and '$f(x)$' for x-label and y-label respectively. Save the plot in a suitable format. 20

7. Plot the following parametric functions for $0 \leq \theta \leq 2\pi$:

$$f_x(\theta) = (1 + \cos\theta)\cos\theta, \; f_y(\theta) = (1 + \cos\theta)\sin\theta.$$

Set the title of the plot as "Parametric plot". Set sample size 1000. Save the plot in a suitable format. 20

Group - B

8. (a) Write a python program to find the solution of transcendental equation $x \ln(x) = x$ using bisection method. Find the solution lying in the range [1, 4] with three decimal places of accuracy.

   (b) Write a python program to calculate the sum of the following A.P. series :

   $S = 1 + 5 + 9 + 13 + \ldots\ldots$ up to 8 terms.

   Print the output. 30+10

9. (a) Write a python program to compute the root of the equation $e^x - \sin(x) = 0$ using Newton-Raphson method with initial value $x = -2$, correct up to three decimal places. Print the root.

   (b) Write a python program to calculate the trace of the following matrix :

$$M = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ -7 & -8 & -9 \end{pmatrix}$$

   Print the output. 30+10

10. (a) A first order differential equation is given as below :

$$y\frac{dy}{dx} = x \text{ with } y(1) = 2.$$

   Create list (say X) with 1000 x-values between [1, 2]. Solve this differential equation using Euler method for x-values in the list X. It is given that the exact solution is

$$y^2 = x^2 + 3$$

   Calculate and print the value of y at $x = 2$. Compare your answer with the value from the exact solution.

   (b) Write a python program to calculate the sum of the following series :

$$S = \frac{1.2}{3.4} + \frac{5.6}{7.8} + \ldots\ldots \text{ up to 4 terms.}$$

   Print the output. 30+10

11. (a) Using Newton-Raphson method with starting point $x_0 = 5$, determine the root of the following equation correct up to three decimal places.

$$x^3 - 7x^2 + 8x - 3 = 0.$$

   Print the root.

   (b) Write a python program to calculate the transpose of the following matrix :

$$M = \begin{pmatrix} 3 & 4 & -2 \\ 1 & -5 & -1 \\ 2 & 6 & 9 \end{pmatrix}.$$

   Print the output. 30+10

12. (a) Acceleration of a particle can be written as

$$v\frac{dv}{dx} = \frac{\sin(x)}{x^2 + 11x + 28}.$$

Write a python program to solve the above differential equation using Euler method and find the velocity at $x = 10$m, provided the initial velocity $v(x = 0) = 1$. Print the output. Take the increment of x as $dx = 0.001$.

(b) Write a python program to find the sum of the following A.P. series :

$S = 7 + 11 + 15 + \ldots\ldots$ up to 6 terms.

Print the output. 30+10

13. (a) Put the following numbers in a list :

2.83, − 8.31, − 3.67, 7.46, − 0.73, 1.39, − 7.38, 5.92.

Write a python program to sort these numbers in ascending order. Store and print the sorted list. Create another list with the squares of the previous (sorted) list. Print this new list. Determine the sum of the squared elements of the list and print it.

(b) Make the following data table in a file. Write a python program to read the columns of this file with two lists say X and Y

−1.0	2.718
−0.8	1.424
−0.6	0.656
−0.4	0.239
−0.2	0.049
0.0	0.0
0.2	0.033

Print the two lists X and Y. 30+10

14. (a) Sort the following integers in ascending order and put them in a list.

11, 28, 17, 5, 19, 15, 7, 31, 23, 13. Print the sorted list.

(b) Determine the prime numbers from the sorted list and print those numbers. 10+30

2023

PHYSICS — HONOURS — PRACTICAL

Paper : SEC-1P

Full Marks : 100

The figures in the margin indicate full marks.

SET - 3

[LNB - 20, VIVA - 20, PROGRAMMING - 60]

Answer ***any one*** question from ***each group.***

Group - A

1. Plot the following functions for $-1 \leq x \leq 1$ with same x-y axes.

$$f(x) = \sinh(0.5x)\cos 2x$$
$$g(x) = 0.2x^2$$
$$h(x) = 0.1\tan x.$$

Put the x-label and y-label as 'x' and '$f(x)$' respectively. Set the title of the plot 'function plots'. Save the plot in a suitable format. 20

2. Write the following data table in a file. Fit the following data-file with the function

x	0.0	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0
y	0.0	0.214	0.216	0.154	0.091	0.045	0.018	0.005	−0.0	−0.002	−0.002

$f(x) = \sin(ax)e^{bx}$ for the parameters a, b. Write the values of a, b. Plot the data file along with $f(x)$. Save the plot in a suitable format. 20

3. Make the following data table in a file. Write a gnuplot script to fit the data with the curve

x	0.0	0.349	0.698	1.047	1.396	1.745	2.094	2.443	2.792	3.142
y	0.0	0.241	0.32	0.304	0.244	0.172	0.107	0.056	0.021	0.0

$f(x) = a\sin(x)e^{bx}$ having parameters a and b. Write the values of a and b. Plot the above data table along with $f(x)$. Save the plot in a suitable format. 20

4. Plot the following polar functions for $-\pi \leq 0 \leq \pi$.

$\cos 20, \sin 20$

Set the plot title 'Example of polar plot'. Set sample 1000. Save the plot in a suitable format. 20

5. Plot the following functions :

$$f(x) = \frac{3\pi}{2}\frac{\sin x}{x} + \cos x$$
$$g(x) = -1$$
$$h(x) = 1$$

with same x-y axes for $-2\pi \leq x \leq 2\pi$ with different line types. Set sample size 1000. Save the plot in a suitable format. 20

6. Plot the following piece-wise function using tarnery operator.

$$f(x) = -5, \quad x < 0$$
$$= 2x - 3, \; 0 \leq x < 2$$
$$= x^2, \quad x \geq 2$$

Put the x-label and y-label as 'x' and '$f(x)$' respectively. Set sample 1000. Save the plot in a suitable format. 20

7. Plot the following parametric functions using GNUPLOT :

$x(t) = 8 \sin t (2\cos^3 t - \cos t)$, $y(t) = 4 \cos 2t$ for $-\pi \leq t \leq \pi$. Set x-label as "t". Set title of the graph as "Parametric Plot". 20

Group - B

8. (a) Write a python program to determine the root of the following equation using Newton-Raphson method correct up to three decimal places of accuracy :

$$x^2 - 8x + 11 = 0$$

Take $x = 6$ as guess value. Print the root.

(b) Define a python function for the following mathematical function :

$$f(x) = x^2 e^{-x^2}.$$

Find the values of $f(x)$ at $x = 1.2$ and $x = 2.1$ and print them. 30+10

9. (a) Write a python program to solve the following differential equation using Euler method :

$$\frac{dy}{dx} = xy, \quad 0 \leq x \leq 2$$

with initial condition $y\,(x = 0) = 1$.

Calculate and print the value of y at $x = 2$. Take the increment of x as $dx = 0.001$.

(b) Write a python program to find the factorial of the given number $n = 7$. Print the output. 30+10

10. (a) Using bisection method determine the root lying between [3, 4] of the following equation up to three decimal places of accuracy :

$$2x^2 + 5 = e^x.$$

Print the root.

(b) Write a python program to find the transpose of the following matrix :

$$A = \begin{pmatrix} 2.1 & 3 & 0 \\ 4 & 3.2 & 1 \\ -1 & 0 & 4.3 \end{pmatrix}.$$

Print the new matrix. 30+10

11. (a) Using bisection method find the root lying between [1, 2] of the following equation up to three decimal places of accuracy :

$$\sin 3x + 3e^{-2x} \sin x - 3e^{-x} \sin 2x - e^{-3x} = 0.$$

Print the root.

(b) Write a python program to find the following sum :

$$S = \sum_{n=1}^{6} \frac{1}{n^2}$$

Print the output. 30+10

12. (a) Write a python program using the Newton-Raphson method to find the root of the equation

$$x^3 + (x + 1)^2 + 4x = 20$$

correct up to third decimal places. The root to be found lies inside the range $-2 < x < 3$. Print the output.

(b) Write a python program to find the trace of the following matrix :

$$A = \begin{pmatrix} 1.7 & 2.8 & 3.9 \\ 4.2 & -5.3 & 2.1 \\ 7.1 & 8.2 & 9.3 \end{pmatrix}$$

Print the value. 30+10

13. (a) Using bisection method determine the root lying between [1.3, 2] of the following equation correct up to three decimal places :

$$e^{3x} - 27x^6 + 27x^4e^x - 9x^2e^{2x} = 0.$$

Print the root.

(b) Write a python program to calculate the sum of the following G.P. series :

$S = 4 + 12 + 36 + \ldots$ up to 5 terms.

Print the output. 30+10

14. (a) Write a python program to take a positive real number x as input from the user. Define a new variable $z = \dfrac{x-1}{x+1}$. Calculate the sum of the following series with an accuracy of 0.001.

$$s = \frac{1}{2}\left(z + \frac{z^3}{3} + \frac{z^5}{5} + \frac{z^7}{7} + \ldots\ldots\right).$$

Print the sum and number of terms where the series is terminated with the aforesaid accuracy. Also print the value of e^s.

(b) Consider the following two matrices :

$$A = \begin{pmatrix} 1.1 & 2.2 \\ 3.3 & 4.4 \end{pmatrix}, \quad B = \begin{pmatrix} -2 & 0 \\ 3 & 1 \end{pmatrix}$$

Write a python program to find $A - B$. Print the resultant matrix. 30+10